

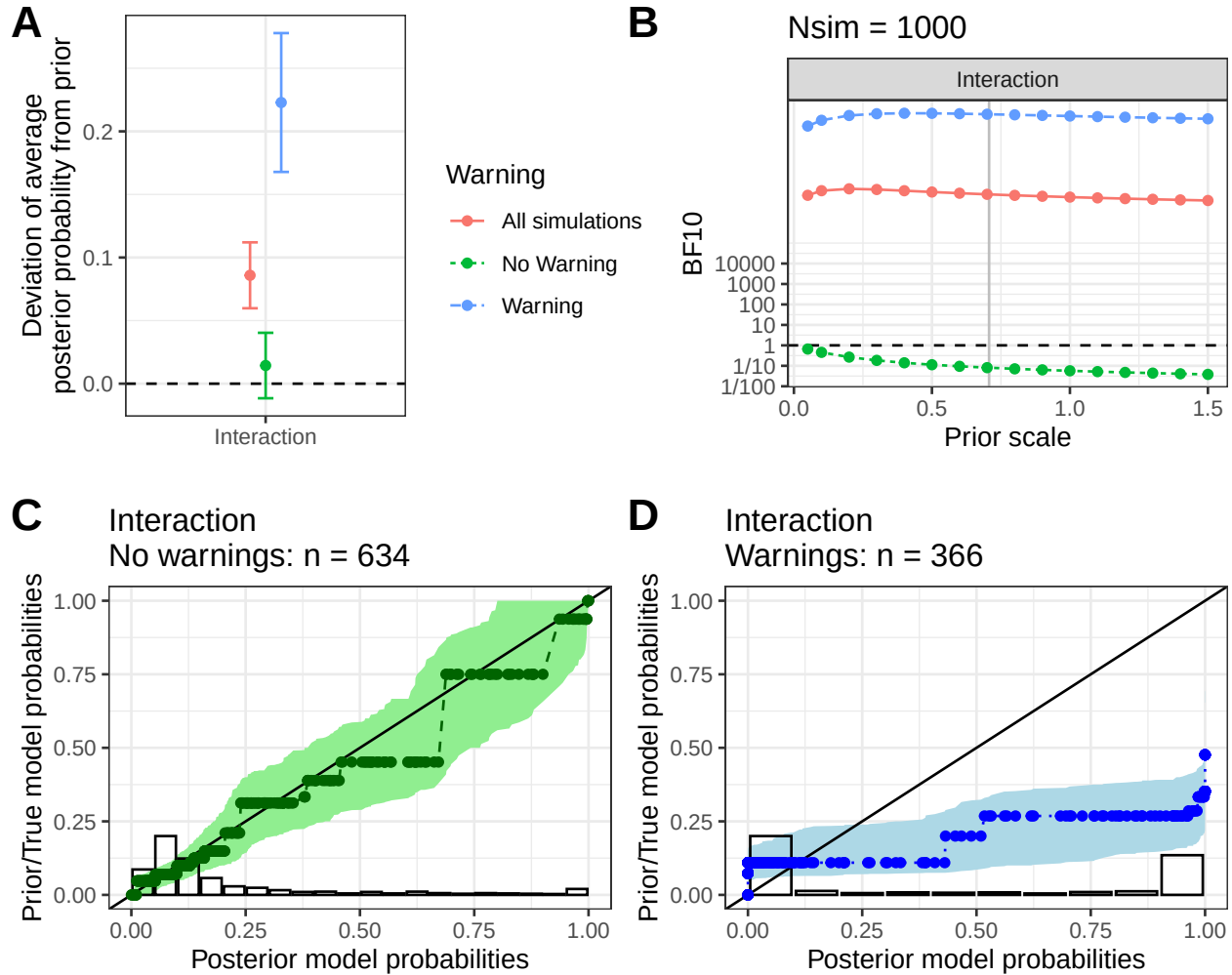


Figure 5. Repetition of the previous analyses of the interaction term (in design 3), but with 1000 SBC simulations and 10,000 iterations of MCMC sampling. A) The average deviation of posterior model probabilities from the prior (95% CI). All simulations and simulations without / with a warning message. The horizontal broken line indicates absence of deviation; deviations from this line indicate bias. B) Bayesian t-tests of whether the posterior probability for H1 deviates from the prior, with sensitivity analyses for the prior scale. The vertical grey line indicates a default Bayes factor. C+D) Reliability diagrams: the prior model probability is plotted as a function of the estimated posterior model probabilities for simulations without (C) and with (D) a warning message. Error bands show 95% consistency bands. Results show that when no warning is issued prior model probabilities do not deviate from posterior model probabilities, suggesting accurate estimation, but that in simulations with a warning message they clearly deviate, providing evidence for bias in this case.

$BF_{01} > 10$ for the null hypothesis, which was evident especially for larger prior scales. These results suggest evidence for absence of overall bias in the Bayes factor estimates when no warning message is issued. However, in the cases where a warning message was issued by the bridge sampler, we saw clear evidence for bias in the Bayes factor estimates, with average posterior model probabilities being too large (and larger than the prior probability; $BF_{10} > 10000$).

Moreover, reliability diagrams (see Figure 5C+D) showed that now - with sufficient SBC simulations - for the simulations where no warning message was shown, prior and posterior model probabilities did not deviate, suggesting no bias in these cases. Moreover, in simulations where a warning message was issued by the bridge sampler, the reliability diagrams clearly showed that posterior and prior model probabilities did not match.

The results with larger numbers of SBC simulations therefore suggest that Bayes factor estimates are indeed accurate in simulations without a warning message, but are biased in simulations with a warning message.

Results as a function of the number of samples. In the previous analyses, we used different numbers of samples of SBC. As noted above, the number of SBC samples could impact the results since a larger number of samples may be needed to obtain enough power to detect divergences between the posterior and prior model probabilities, and to yield stable and reliable results. To investigate this issue, we next studied the results as a function of the number of SBC samples, i.e., performing a re-analysis of the previous marginal SBC simulations.

The results showed (see Fig. 6) that evidence for bias in simulations with warning messages was quickly evident with already less than 100 simulation runs, and became more and more clear with increasing runs. Evidence for accurate Bayes factor estimates in simulations without warning messages accumulated less quickly and remained at less clear levels with a Bayes factor of around $BF_{01} = 10$.

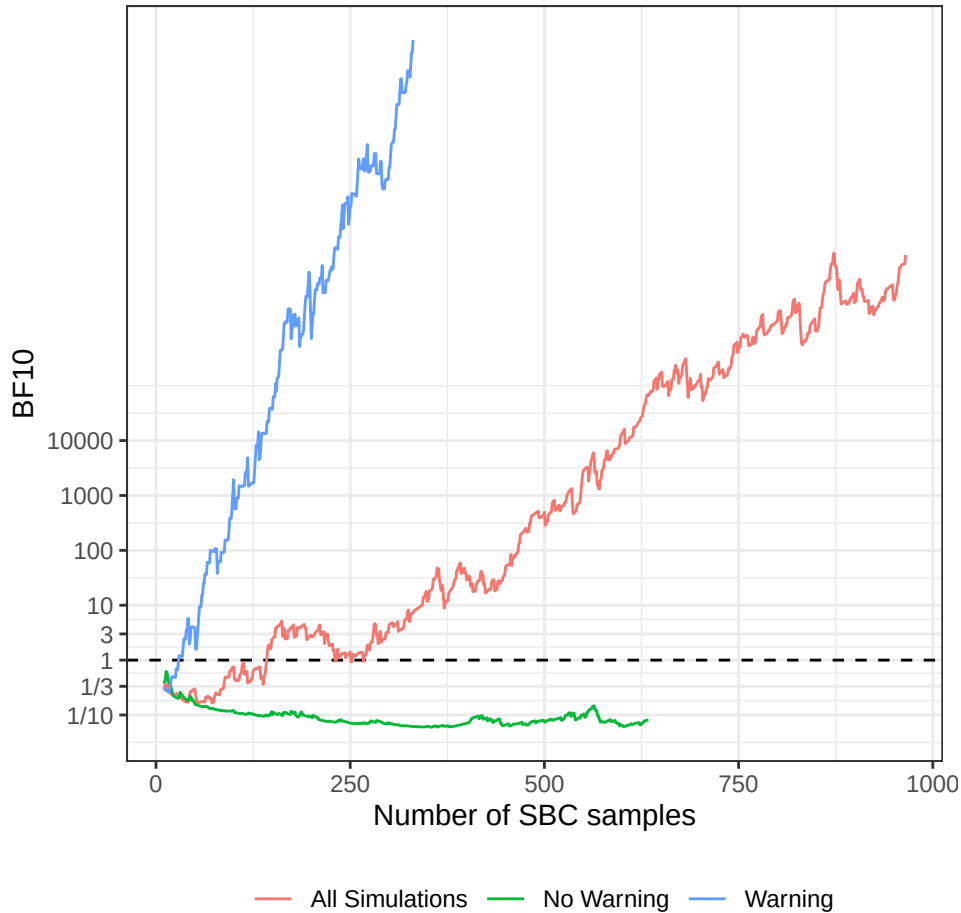


Figure 6. Analysis of how the number of SBC samples impacts marginal SBC. The result of a Bayesian default t-test of whether the average posterior model probability diverges from zero is plotted as a function of the number of SBC samples used in the analysis. Results are shown for all simulations (red) and for simulations without (green) and with (blue) a warning message. Evidence for bias in simulations with warning messages is visible very quickly, with only few SBC simulation runs, whereas evidence for accuracy without warning messages accumulates more slowly.